STATS 201A Lecture 6

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$$Y_{ij} = N(\mu_i, \sigma_i^2)$$
$$\hat{\mu}_i = \bar{y}_{i\cdot}$$

Split plot design

$$\begin{aligned}
\mathrm{Var}(\mathrm{ME}(A)) &= \mathrm{Var}\left(\bar{z}(A+) - \bar{z}(A-)\right) \\
&= \frac{\sigma^2}{8} + \frac{\sigma^2}{8} \\
\mathrm{Var}(y_{ij}) &= \sigma^2 \\
\mathrm{ME}(B \backslash A-) &= y_B - y_A \\
\mathrm{INT}(A, B) &= \frac{1}{2}\left\{\mathrm{ME}(B \mid A+) - \mathrm{ME}(B \mid A-)\right\}
\end{aligned}$$